

# Output Content (OC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MATHDIAL | Context: Indian Metropolitan Professional Teachers (Grades 1–8)

**Output Content definition:** Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

User-flagged HIGH risk and the highest-priority validity concern. Ground-truth quality judgments were produced by 91 professional teachers recruited via Prolific [\[Q23\]](#), predominantly UK nationals, followed by USA, Canada, Australia, India, Germany [\[Q27\]](#) — India is the fifth of six named countries with no disaggregated count reported. Annotators were explicitly instructed not to correct student solutions by telling but to give students opportunities to explore [\[Q148\]](#), systematically encoding Western constructivist norms regardless of annotator origin. Inter-annotator agreement is concerning:  $\kappa=0.60$  between independent annotators but only  $\kappa=0.49$  and  $\kappa=0.34$  between annotators and original teacher for move labeling [\[Q69\]](#), and critically  $\kappa=0.34$  for Equitable Tutoring [\[Q100\]](#) — the dimension most sensitive to cross-cultural variation. The user explicitly expects systematic divergence between Indian professional teachers' judgments and the benchmark's annotator-derived ground truth, particularly on direct-correction vs. Socratic probing and exam-readiness weighting [\[A4\]](#). Empirically, the dataset shows pervasive Socratic-mode dialogues with multiple Focus/Probing moves before any Telling [\[DATASET-D2, DATASET-D3\]](#), gentle non-directive corrections [\[DATASET-D20\]](#), and 'had to reveal' framing on successful Telling resolutions [\[DATASET-D5, DATASET-D23\]](#) — patterns the user identifies as misaligned with Indian professional norms. Web research corroborates: PARIKSHA found LLM evaluators align less with human evaluations on culturally nuanced responses [\[WEB-18\]](#), and systematic review confirms majority Indic datasets are translations with poor socio-cultural alignment [\[WEB-15\]](#). No India-specific math tutoring dialogue benchmark exists [\[WEB-15, WEB-16, WEB-17, WEB-18\]](#). This violates both convergent validity (labels do not correlate with regional perspectives) and external validity (judgments do not generalize).

| ID                   | Evidence                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Q23</a>  | "We recruit professionals with teaching experience through Prolific. We only select teachers who have completed at least 500 submissions and achieved a 100% completion rate." (p.4)                                                                                                                                                                                           |
| <a href="#">Q27</a>  | "The majority of annotators are nationals of the UK, followed by the USA, Canada, Australia, India, and Germany, with a median age of 39 years." (p.4)                                                                                                                                                                                                                         |
| <a href="#">Q69</a>  | "We obtain an agreement of $\kappa = 0.60$ between the two annotators and $\kappa = 0.49$ and $\kappa = 0.34$ , respectively, between either of the annotators and the teacher." (p.6)                                                                                                                                                                                         |
| <a href="#">Q100</a> | "We obtain agreements of $\kappa = 0.29$ , $\kappa = 0.69$ , and $\kappa = 0.34$ for the three categories." (p.8)                                                                                                                                                                                                                                                              |
| <a href="#">Q148</a> | "Specifically, they were instructed to not just correct student solutions by telling what's correct/incorrect, but to give students the opportunity to explore the problem with a focus on core aspects, such as their chosen strategy." (p.17)                                                                                                                                |
| <a href="#">D2</a>   | Hi Rishi. Can you tell me how many hours it would take Apple to run 24 miles?" ... "Good. How many hours would it take Mac?" ... "How much longer does it take Apple?" ... "How much longer is that in minutes?" ... "So how much faster is Mac than Apple in minutes?" — Six consecutive probing questions guiding student to answer; entirely Socratic, no direct correction |
| <a href="#">D3</a>   | Carmela has $600 \times 2 + 50 = 1250$ " ... "It is 1250. Not 2450" ... " $600+950+1250 = 2800$ — Teacher resorts to Telling only after multiple Focus moves fail; Telling framed as reluctant fallback                                                                                                                                                                        |
| <a href="#">D5</a>   | No $32-7 = 25$ . So the difference is 25" ... "No. His total running time was 25 seconds. You need to subtract the amount of time it took him to run up the first half of the hill to find your answer — Telling moves are labeled and visible in transcript; student resolves confusion only after explicit revelation                                                        |
| <a href="#">D20</a>  | Teacher: I feel you have made this more complicated than needed could you maybe find a simpler way — Generic, mild correction without directing student to specific error; low-directive style                                                                                                                                                                                 |
| <a href="#">D23</a>  | Teacher: (telling)You need to find out how many times 48 goes into 192" / "Teacher: (telling)192 divide by 48 =" / "Teacher: (telling)4 is the correct answer you dont need to calculate further — Three consecutive Telling moves before student accepts correct answer; demonstrates reluctant resort to direct instruction                                                  |

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|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>WEB-15</b> | Systematic review: majority Indic LLM datasets are translations and models fare poorly on socio-cultural tasks ( <a href="https://arxiv.org/html/2501.13912v1">https://arxiv.org/html/2501.13912v1</a> ) |
| <b>WEB-16</b> | IndicEval covers Indian high-stakes exams but is not a pedagogical dialogue benchmark ( <a href="https://arxiv.org/html/2602.16467">https://arxiv.org/html/2602.16467</a> )                              |
| <b>WEB-17</b> | Hindi LLM benchmarks suite focuses on math reasoning not tutoring quality ( <a href="https://arxiv.org/abs/2508.19831">https://arxiv.org/abs/2508.19831</a> )                                            |
| <b>WEB-18</b> | PARIKSHA finding: LLM evaluators align less with human evaluations on culturally nuanced responses ( <a href="https://arxiv.org/html/2406.15053v1">https://arxiv.org/html/2406.15053v1</a> )             |

## Strengths

- Annotators are professional teachers with vetted credentials (500+ submissions, 100% completion, training tests) [Q23, Q24, Q25, Q132]
- India represented at all in the annotator pool [Q27]
- Authors transparently report inter-annotator agreement statistics including low agreement on Equitable Tutoring [Q100]
- Self-correctness annotations enable filtering by resolution strategy [DATASET-D5]

| ID          | Evidence                                                                                                                                                                                                                                                                                                             |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q23</b>  | "We recruit professionals with teaching experience through Prolific. We only select teachers who have completed at least 500 submissions and achieved a 100% completion rate." (p.4)                                                                                                                                 |
| <b>Q24</b>  | "Annotators read guidelines for the task in an initial training phase (cf. Section D.3) and then complete a test on an example conversation to assess their understanding of the task." (p.4)                                                                                                                        |
| <b>Q25</b>  | "We only select annotators with 100% test scores for further rounds of data collection, similar to Zhang et al. (2023)." (p.4)                                                                                                                                                                                       |
| <b>Q27</b>  | "The majority of annotators are nationals of the UK, followed by the USA, Canada, Australia, India, and Germany, with a median age of 39 years." (p.4)                                                                                                                                                               |
| <b>Q100</b> | "We obtain agreements of $\kappa = 0.29$ , $\kappa = 0.69$ , and $\kappa = 0.34$ for the three categories." (p.8)                                                                                                                                                                                                    |
| <b>Q132</b> | "To ensure the data quality we filter only annotators with 100% completion rate with more than 500 total submissions." (p.16)                                                                                                                                                                                        |
| <b>D5</b>   | No 32-7 = 25. So the difference is 25" ... "No. His total running time was 25 seconds. You need to subtract the amount of time it took him to run up the first half of the hill to find your answer — Telling moves are labeled and visible in transcript; student resolves confusion only after explicit revelation |

## Checklist

### Checklist item definitions:

| ID          | Assessment Question                                               |
|-------------|-------------------------------------------------------------------|
| <b>OC-1</b> | Do ground-truth labels reflect regional stakeholder perspectives? |
| <b>OC-2</b> | Assess potential annotator-regional population disagreement       |
| <b>OC-3</b> | Review annotator demographics from documentation                  |
| <b>OC-4</b> | Consider label re-annotation by regional annotators               |
| <b>OC-5</b> | Review aggregation methods for minority perspective erasure       |
| <b>OC-6</b> | Document label issues harming convergent/external validity        |

### Checklist responses:

**OC-1:** No — ground-truth labels reflect predominantly UK/US pedagogical perspectives [Q27] with annotators explicitly instructed toward Socratic exploration [Q148], not Indian metro professional teacher perspectives.

**OC-2:** Yes — substantial disagreement is expected. The user explicitly anticipates systematic divergence on direct-correction vs. Socratic probing and exam-readiness weighting [A4]. PARIKSHA documents LLM-human alignment gaps on culturally nuanced responses [WEB-18].

**OC-3:** Annotator demographics partially documented: 91 teachers, 71 female/18 male, median age 39, predominantly UK nationals [Q26, Q27]. Indian representation is not disaggregated [annotator\_provenance gap].

**OC-4:** Re-annotation by Indian metro professional teachers is strongly warranted given user-anticipated systematic divergence [A4].

**OC-5:** Aggregation methods ( $\kappa$  for inter-annotator agreement) [Q69, Q100] do not separately surface minority-country perspectives. With Indian annotators a small minority, their judgments would be erased in any majority-aggregation procedure.

**OC-6:** Convergent validity violation: Western-norm labels unlikely to correlate with Indian teacher judgments [A4, WEB-18].  
External validity violation: original judgments will not generalize to Indian metro context [WEB-15].

| ID     | Evidence                                                                                                                                                                                                                                        |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q26    | "We employ 91 expert annotators, of which 71 identify as female and 18 as male." (p.4)                                                                                                                                                          |
| Q27    | "The majority of annotators are nationals of the UK, followed by the USA, Canada, Australia, India, and Germany, with a median age of 39 years." (p.4)                                                                                          |
| Q69    | "We obtain an agreement of $\kappa = 0.60$ between the two annotators and $\kappa = 0.49$ and $\kappa = 0.34$ , respectively, between either of the annotators and the teacher." (p.6)                                                          |
| Q100   | "We obtain agreements of $\kappa = 0.29$ , $\kappa = 0.69$ , and $\kappa = 0.34$ for the three categories." (p.8)                                                                                                                               |
| Q148   | "Specifically, they were instructed to not just correct student solutions by telling what's correct/incorrect, but to give students the opportunity to explore the problem with a focus on core aspects, such as their chosen strategy." (p.17) |
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## Information Gaps

- Disaggregated count of Indian annotators in the pool [annotator\_provenance unresolvable from public sources]
- Whether any sampled dialogues were authored by Indian-origin annotators (dataset does not expose annotator identity per dialogue)
- Magnitude of expected  $\kappa$  between Indian metro teachers and benchmark ground truth on Equitable Tutoring

## Requires Expert Verification

- Whether re-annotation of a representative subsample by Indian metro teachers produces label distributions that diverge significantly from MATHDIAL ground truth on Equitable Tutoring
- Whether Indian teachers would re-rate currently-low-rated directive dialogues as high-quality

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## Remediation

**Gap 1: Ground-truth quality labels are dominated by UK/US annotators explicitly instructed toward Socratic non-direction, with  $\kappa=0.34$  on Equitable Tutoring even within the Western pool. Systematic divergence from Indian professional teachers' judgments expected.**

**Recommendation:** Re-annotate a stratified sample ( $\geq 100$  dialogues, balanced across self-correctness outcomes) with at least 5–10 Indian metro professional teachers per item. Compute  $\kappa$  between Indian and original annotators per dimension; report dimension-level disagreement and use the Indian-annotated subset as the operative ground truth for deployment evaluation.

**Gap 2: Dataset does not expose annotator identity per dialogue, so it is impossible to filter to dialogues authored by Indian-origin annotators or audit demographic representativeness at the item level.**

**Recommendation:** Request annotator-disaggregated metadata from the benchmark authors, or treat the entire dataset as Western-norm-anchored and weight evaluation results accordingly. Flag the unknown Indian-annotator contribution as a residual risk in deployment documentation.